

C Research & Development

In the pharmaceutical industry, the current R&D model is facing considerable challenges.¹ Since the 1990s, new product approvals have been relatively rare while costs have continued to rise.² With mature markets stagnating, companies are looking elsewhere for growth. There is huge demand for R&D that targets the needs of people living in developing countries: needs for new products for certain diseases, and for adapted versions of existing products for specific populations and conditions. However, the pharmaceutical markets in many of these countries are fragile. As companies look to develop products with viable markets, it seems likely that certain diseases, conditions and patient groups will continue to be overlooked.

In this chapter, the Index reports on how companies are meeting the demand for new and adapted products in these countries through research and development. It also looks at how companies are engaging in new partnership models that aim to pool the risks associated with R&D.³

The Index examines how company pipelines target 47 high-burden diseases, grouped into four categories: communicable diseases; non-communicable diseases; neglected tropical diseases, and maternal & neonatal health conditions. Compared with 2012, the 2014 Index raises the bar, incorporating new areas of investigation, demanding higher standards of evidence and using more stringent scoring guidelines.

Four areas of analysis

Product development

Reporting on companies' approaches to product development for diseases covered by the Index.

Collaborations and IP sharing

Reporting on whether and how companies engage in R&D partnerships, and whether the underlying agreements are based on access-oriented terms and conditions.

Clinical trial conduct and data transparency

Reporting on clinical trial codes of conduct, evidence of oversight and enforcement mechanisms for these codes and on breaches of clinical trial conduct in countries within scope. It also examines transparency around clinical trials and whether companies share clinical trial data.

Innovation in R&D

Reporting any innovative, sustainable or open business model that addresses current gaps in product development for diseases within scope.

Top findings in Research & Development

Evidence of sustained commitment to R&D for relevant diseases

Most companies have an R&D strategy in place that explicitly takes patients in developing countries into account. The Index captured 327 R&D products that target high-burden diseases for countries within scope, a large proportion of which are new to the 2014 Index. Most companies have also successfully moved products along their pipelines:

11 companies have been granted regulatory approvals for 30 new products, collectively covering 11 of the 47 conditions within scope.

R&D is mainly focused on five diseases

More than half of all products in development (54%) target five diseases: diabetes, lower respiratory infections, hepatitis, HIV/AIDS and malaria. These same diseases also have the most products in clinical development. The most notable gaps are for maternal and neonatal health conditions: there is a lack of products being adapted for safe and effective use by pregnant women and neonates. The majority of products for neglected tropical diseases are early drug-discovery projects or pre-clinical studies. It will be some time before any of these products are on the market.

Companies devote large pipeline shares to relevant diseases

Almost half of companies are devoting significant shares of their pipelines to diseases within scope for patients living in countries within scope. For three companies this reaches over 35%. Another six companies devote more than 20% of their pipeline to relevant R&D projects. In total, the industry is developing more than 300 products for relevant diseases.

Substantial proportion of R&D takes place in access-oriented partnerships

Approximately 40% of all product development occurs in collaboration, with either publicly funded, privately funded or a mix of partners. More than a third (39%) of these product development partnerships are based on research contracts that include access provisions. Since the 2012 Index, the total number of R&D partnerships has increased by 35%, mainly due to an increase in drug-discovery and early-stage partnerships targeting neglected tropical diseases, malaria and tuberculosis.

Limited consideration of access to compounds for non-communicable diseases

Most companies struggle to demonstrate how innovative compounds targeting non-communicable diseases will be accessible to patients living in countries within scope. While there are several such compounds in clinical development, there is no implementation strategy that addresses these patients. No company makes commitments to registering such products in relevant countries. The terms and conditions of partnerships for non-communicable product development are not disclosed to the Index or to the public.

Movement towards stronger ethical clinical trial standards and enforcement

More companies provided strong evidence of having enforcement measures in place to ensure ethical clinical trial conduct for in-house and outsourced trials that are in line with guidelines on good clinical practice (ICH-GCP). This has increased from four companies in 2012 to ten. Seven companies incorporate aspects specific to the Declaration of Helsinki in their clinical trial codes of conduct.

How the companies perform

Companies receive a wide spread of scores in this area, driven largely by their performances in product development. Companies with the strongest relevant pipelines also have strong R&D strategies that are based on meeting the needs of patients in countries within scope. Once again, the industry is developing a substantial number of products for high-burden diseases, most of which are in clinical development.

In R&D, the companies are divided into three distinct groups. There are two companies in the lead, separated by a small margin – GSK and Johnson & Johnson. A pack of three companies follow close behind – Merck KGaA, AbbVie and Novartis. The following ten companies comprise the second group. Although the scores in this group decrease gradually, there is a major difference between the highest- and lowest-ranked company. Another five companies lag behind.

Leaders score highly across the board

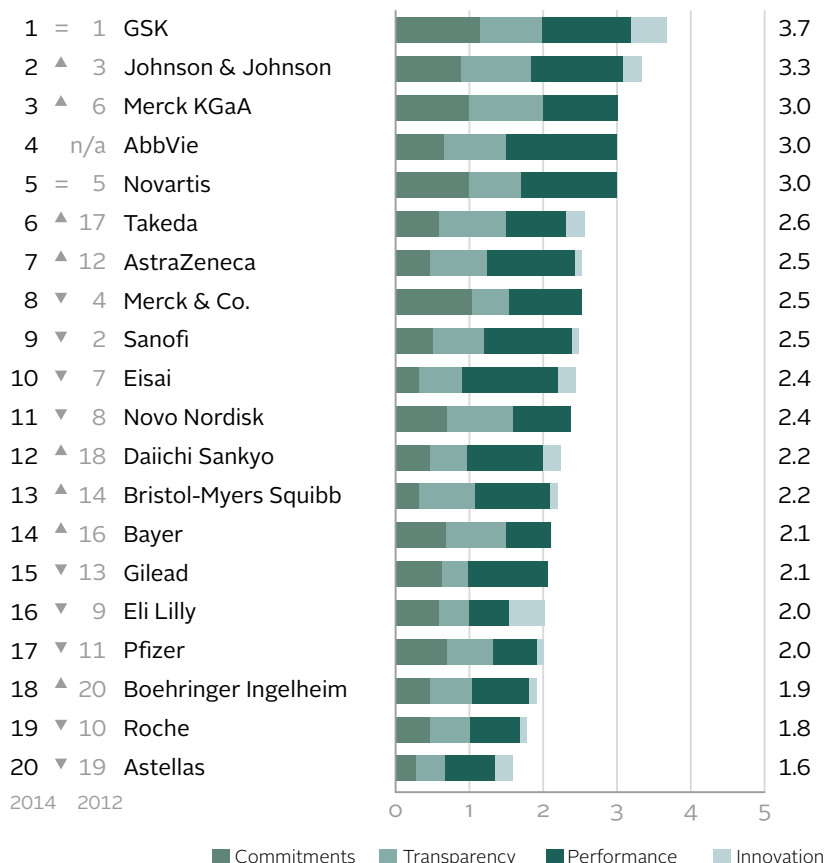
All the top five companies in R&D score highly for product development, measured across several parameters, including R&D strategies, R&D investments and how much of their overall pipeline is devoted to developing products relevant to the Index.

In 1st place, GSK remains the overall leader for R&D. As well as scoring highly for product development, it scores the highest for collaborations, IP sharing and innovation. It has a far-reaching R&D strategy, which specifically addresses unmet needs relating to neglected diseases, anti-microbial resistance and non-communicable diseases in Africa.

In 2nd place, Johnson & Johnson leads in product development, with an exten-

Figure 25

Company Ranking Research & Development



sive relevant pipeline that includes numerous adapted formulations, including fixed-dose combinations and paediatric formulations. Compared to its peers, it has moved the most products through its pipeline phases. Tied in 3rd place, Merck KGaA, AbbVie and Novartis have strong scores across the board. Merck KGaA focuses on developing medicines for non-communicable diseases. Its R&D strategy includes operationalisation strategies for relevant countries that cover multiple diseases, including diabetes and asthma. AbbVie also has a strong position in product development relating to the Index disease scope, including products inherited

from Abbott. Novartis has the largest relevant pipeline of all companies, with a strong focus on malaria, vaccines and maternal and neonatal health conditions.

The leaders are developing many of their products on access-oriented terms, as shown in the terms of their product development partnerships. This illustrates a strong level of commitment to tackling diseases with the highest burdens in countries with the poorest patients.

The leaders also go beyond compliance with the International Conference on Harmonisation guideline for Good

Clinical Practice (ICH-GCP)⁴ in their codes of conduct for clinical trials, and are moving towards compliance with the Declaration of Helsinki.⁵ Their codes of conduct include ethical study-design considerations, post-trial access to drug candidates and provisions for offering compensation following serious adverse events.

Partnerships push newcomers into middle group

There is a large middle group of ten companies with varying performances across all parameters. AstraZeneca and Takeda have risen to the top of this group since 2012, and both engage in a substantial number of R&D partnerships. Merck & Co. follows, with little movement along its relevant pipeline since 2012. Sanofi has a large number of relevant investigational products. However, when engaging in product development partnerships, it falls behind when it comes to taking access into account.

Eisai, Bristol-Myers Squibb and Gilead also have large relative pipelines, but do not meet the same standards as the leaders for clinical trial conduct and transparency. Novo Nordisk has strong ethical codes of conduct and procedures in place for its clinical trials, yet a small relevant pipeline. Daiichi Sankyo and Bayer both provide evidence of

having access-oriented R&D partnerships, yet also have small relevant pipelines.

Laggards have smaller pipelines

Eli Lilly, Pfizer, Boehringer Ingelheim, Roche and Astellas occupy the bottom of the ranking. For these companies, only a small share of their pipelines target diseases and countries within the scope of the Index, and they provide little evidence of taking access into account during R&D. Boehringer Ingelheim stands out from the other laggards for rising two places since 2012.

Biggest risers: Takeda, Daiichi Sankyo, AstraZeneca

Takeda and AstraZeneca climb 11 and five places respectively. Both engage in numerous IP-sharing partnerships and incorporate access provisions into a large proportion of their R&D partnerships. Takeda is involved in a product development partnership with the Medicines for Malaria Venture that has reached the clinical stage of development. During the period of analysis, AstraZeneca shared IP with numerous partners through the WIPO Re:Search initiative.³ AstraZeneca and Takeda comply with the WHO's standards for registering trials and publishing results, and consider sharing patient-level data on a case-by-case basis.

Daiichi Sankyo has also risen considerably, from 18th to 11th place. This is partly because, for the first time, it disclosed to the Index the relevant investigational products it obtained via the acquisition of Ranbaxy. Daiichi Sankyo also shares proprietary compounds for testing via the Global Health Innovative Technology Fund (GHIT).

Furthest fallers: Roche, Eli Lilly, Sanofi

Roche and Eli Lilly both fall from the middle of the ranking to the bottom (falling nine and seven places, respectively) due to low scores across the board. In addition, relatively small proportions of their pipelines qualify for analysis.

Sanofi also dropped seven places, despite its large pipeline. It is less transparent than the leaders about the terms and conditions of its product-development partnerships, and it does not explicitly commit to including access-oriented terms. Additionally, it did not disclose its criteria for selecting third parties when outsourcing clinical trials. Sanofi was also pushed out of the top five and into the middle group by strong performances from Merck KGaA, AbbVie, AstraZeneca and Takeda.

INDUSTRY

Evidence of sustained commitment to R&D for relevant diseases

Large shares of companies' R&D pipelines target conditions covered by the Index – over 35% in several cases. In total, companies are developing 327 products that target diseases relevant to the Index. Most products in development target infectious diseases. Communicable diseases account for 47% of the pipeline, followed by non-communicable diseases (36%), neglected tropical diseases (13%), and maternal and neonatal health conditions (4%). It is promising that approximately 64% of all investigational products included in the Index 2014

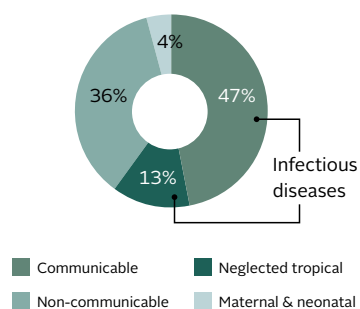
pipeline are medicines and vaccines being tested in clinical trials.

Since 2012, there has been a lot of movement in the industry pipeline, illustrating the industry's ongoing commitment to the development of innovative and adaptive products. Looking only at the scope of the 2012 Index, 42% of pipeline projects captured in 2014 are new. Most target communicable diseases, followed by non-communicable diseases. 33% of pipeline products captured by the 2014 Index were also captured in 2012.

Other projects have been discontinued since 2012, possibly due to compound failure or divestment. The proportion of the pipeline that is due to the expansion of the Index disease scope and pipeline inclusion criteria accounts for 25% of the industry pipeline.

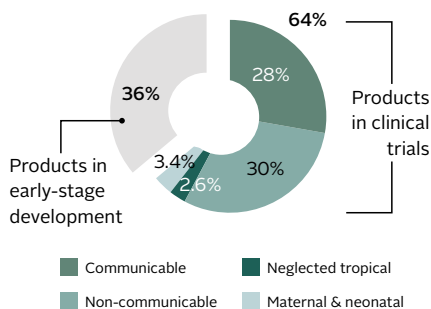
Most companies have also successfully moved products along the pipeline. Eleven companies have been granted regulatory approvals for 30 products, collectively covering 11 of the 47 conditions in scope.

Figure 26

Most products are being developed for infectious diseases

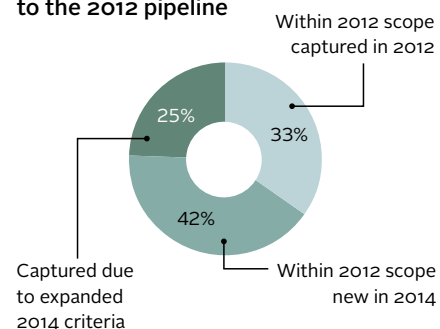
The majority of all R&D products are being developed for infectious diseases (communicable diseases and neglected tropical diseases combined), followed by non-communicable diseases.

Figure 27

Products in development, by disease category and phase

The majority of R&D products are in clinical trials, indicating that the risk of failure is relatively low.

Figure 28

Comparing the 2014 pipeline to the 2012 pipeline

The 2014 pipeline shows substantial movement compared to the 2012 pipeline. Many new projects have been newly added, while others qualify for inclusion this Index.

Companies generally take patients in relevant countries into account when developing products for communicable diseases and neglected tropical diseases. However, most struggled to demonstrate how innovative compounds for non-communicable diseases will reach these patients.

In total, 40% of products in this analysis are being developed in partnerships. The majority of companies engage in product development partnerships. For seven of them, this covers over 10% of their total pipelines. Companies are also collectively more transparent about the terms and conditions attached, revealing whether they contain pro-access provisions, such as price caps or non-exclusive licensing rights in specified territories.

Approximately 75% of companies are engaged in at least one partnership on access-oriented terms. Six companies have signed access-oriented agreements for more than half of all their relevant partnerships, including product development partnerships and IP-sharing partnerships. These typically involve publicly funded organisations, and mainly target malaria, tuberculosis or neglected tropical diseases. It remains unclear whether partnerships with privately funded organisations include access provisions, as companies remain bound by confidentiality agreements. Nevertheless, with so many companies agreeing to access provisions, there is certainly potential for them to move in this direction when engaging with private partners.

Most companies set high standards in their codes of conduct for clinical trials. Going further, almost one-third have ethical aspects from the Declaration of Helsinki⁵ in their codes of conduct. Codes of conduct generally apply to all trials, in-house and outsourced. However, for outsourced trials, only half of companies provided evidence of having robust auditing and monitoring procedures.

Since 2012, companies have improved significantly in sharing clinical trial data. Many companies have implemented systems for sharing patient-level data with trusted partners. The European Medicines Agency's new policy on data sharing has contributed to this trend.

Product development

R&D is mainly focused on five diseases

Why this matters

There is huge demand for R&D that targets the needs of people living in developing countries: both for new products for certain diseases, and for new versions of existing products that have been adapted for specific populations, different age groups and local conditions.

For each company, the Index determines what proportion of its entire R&D pipeline targets diseases within the scope of

the Index^b, firstly with new compounds and products, and secondly, with new versions of existing products, adapted to local populations and conditions. The Index also looks at how frequently the company drives products through the different phases of its R&D pipeline. It checks whether companies' R&D strategies take relevant health priorities into account and are supported by meaningful targets, sufficient investment and other resources.

All products that target communicable diseases and neglected tropical diseases are included in this analysis. For new products targeting non-communicable diseases and maternal and neonatal health conditions, however, the Index looks for additional evidence that companies specifically target R&D efforts towards people living in relevant countries, for example through registration targets or patent and pricing strategies.

How the companies perform

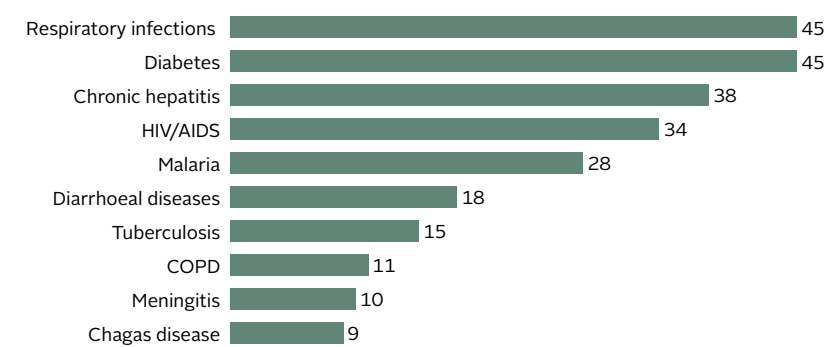
In total, the companies covered by the Index are developing 327 products that qualify for analysis, of which, some are being developed for multiple diseases. More than half of these (54%) target five diseases: diabetes, lower respiratory infections, hepatitis, HIV/AIDS and malaria.

Since 2012, companies have gained regulatory approval for a variety of products targeting 11 diseases or conditions: diabetes, hepatitis, HIV/AIDS, lower respiratory infections, chronic obstructive pulmonary disorder, epilepsy, schizophrenia, meningitis, pertussis, tetanus and tuberculosis.

Companies devote large shares of their pipelines to meeting the needs of patients with diseases in scope and living in countries within scope. For some of the leaders, this reaches as much as 35% of the investigational products in their overall pipeline.

Looking closer, these relevant shares vary when it comes to how and where companies focus their efforts. AbbVie leads when it comes to developing new compounds and products, dedicating more than 25% of its overall pipeline to

Figure 29
Top 10 diseases targeted by the overall industry pipeline



Collectively, the industry is developing 327 products for diseases within scope, including all stages of development. More than half (54%) target the top 5 diseases.

this activity. Johnson & Johnson leads when it comes to adapting existing products, dedicating 25% of its overall pipeline to this. In general, within scope, more companies focus on developing new products than on adapting existing ones to meet local needs.

Comprehensive access-focused R&D strategies still not common
Most companies have an R&D strategy in place that, to an extent, targets diseases within scope. However, only

eight have an R&D strategy that specifically targets multiple disease areas and is supported by specific operationalisation strategies.

More than half of companies provided aggregate figures for their investments in R&D for diseases the Index covers. However, they calculate these figures in various ways. Without an industry-wide standard for such calculations, it remains difficult to assess how companies prioritise relevant R&D activities.

With greater transparency and standardisation, the wider R&D community would be better able to identify and target funding gaps.

Substantial R&D efforts for high-burden communicable diseases

Most companies are developing multiple products for communicable diseases, accounting for the largest proportion (47%) of products included in this analysis. Almost all of these products target just six conditions: HIV/AIDS, malaria, lower respiratory infections, diarrhoeal diseases, tuberculosis and meningitis. Of all products that target communicable diseases, 40% are in early-stage development (discovery and pre-clinical phases).

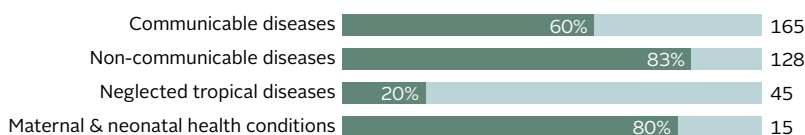
Non-communicable diseases lack implementation strategies

Most companies are also developing at least one product for a non-communicable disease. In total, 36% of all products included in this analysis target this group of diseases, 65% of which target diabetes and hepatitis, of which the latter is linked to cirrhosis of the liver. Conversely, osteoarthritis, epilepsy and nephritis and nephrosis receive little attention. However, for many innovative compounds that target non-communicable diseases, there is no clear strategy for ensuring that they will reach patients living in developing countries. Considering this gap, it is unsurprising that no company has registration commitments for countries within scope for products that target non-communicable diseases.^c

Three companies were able to demonstrate how their R&D efforts for schizophrenia, bipolar affective disorder and unipolar depressive disorders would be accessible to patients living in countries in scope, accounting for only 2% of the overall pipeline. This is of particular concern considering the high prevalence of these diseases globally and the general lack of understanding of local needs.

Figure 30

Proportion of products in clinical trials, by disease category



Compounds can be used for multiple diseases, referred to as multiple indications. Products captured by the Index are being tested for 353 indications in total.

■ Clinical trials (medicines & vaccines)
■ Early stage & technology development

Substantial early-stage development for neglected tropical diseases

Twelve companies are involved in product development for the most prevalent neglected tropical diseases (NTDs); in total, 13% of all products included in this analysis target NTDs and are mostly in early-stage development. Promisingly, this includes nine medicines and vaccines in the clinical stage of development that target dengue, rabies, Chagas disease and trypanosomiasis.

Pregnant women and newborns overlooked

Companies collectively are investing marginal efforts in maternal and neonatal health conditions. Only four companies are active in this area:

- Novartis is the only company that is developing products that target maternal sepsis, maternal haemorrhage and hypertensive disorders of pregnancy.
- Novartis and GSK are developing products specifically for neonatal health: to prevent prematurity and infections through maternal immunisation or by prolonging pregnancy.
- AbbVie is the only company that is developing a medicine that is intended to be administered to premature infants directly (for lower respiratory infections).
- Merck & Co. is developing contraceptives.

Demand for product development remains. Research and development of products adapted for pregnant women and neonates is needed in order to

reduce the enormous disease burden on these groups in low income and middle income countries.

Leaders: strong pipelines, clear strategies

The leaders in product development are Johnson & Johnson, AbbVie, Merck KGaA, Novartis, GSK and Sanofi. Their scores are very close to each other, and as a group, they distinguish themselves by dedicating large proportions of their R&D pipelines to patients with relevant diseases in relevant countries. Together, their six pipelines account for 57% of all relevant products being developed.

These strong pipelines are supported by R&D strategies that are clearly linked to global health priorities and supported by robust operationalisation schemes for achieving set targets. This includes having dedicated units that oversee product development for neglected diseases, engage in partnerships and define targets for registration and investment.

Except Novartis, these companies also specify aggregate amounts of R&D investments for most of the diseases covered by the Index.

Compared to other companies in the Index, Johnson & Johnson dedicates the largest share of its overall pipeline to adapting existing products to meet health needs in relevant countries. It also successfully drove the highest number of investigational products from one phase of the pipeline to the

next during the period of analysis, including gaining regulatory approval for HIV/AIDS, tuberculosis, diabetes and hepatitis medicines. Together, these factors push it into the lead in product development.

Close behind, AbbVie dedicates the largest share of its overall pipeline to developing innovative products that target diseases within the Index’s scope. GSK, Sanofi and Novartis also dedicate large proportions of their pipelines to innovative R&D products.

Merck KGaA has fewer relevant investigational products than other leaders, yet they account for a significant proportion of its total pipeline. The company also publicly discloses its pipeline beyond the clinical phase, and it was the only company to specify its R&D investments for all diseases within the scope of the Index.

Novartis performs well across all parameters, and of all companies, it has the highest total number of relevant

products in its pipeline. However, it does not provide detailed information about its relevant R&D investments.

The middle group: large pipelines but weaker strategies

Following this group are Bristol-Myers Squibb, Gilead, Eisai, Merck & Co. and AstraZeneca. These five companies also dedicate large shares of their pipelines to developing relevant products. They have potential to strengthen their R&D strategies by including, for example, specific time-bound targets and operationalisation strategies. In addition, Gilead and Merck & Co. did not provide evidence of how much they invest in R&D for diseases within scope. Novo Nordisk, Bayer, Daiichi Sankyo, Takeda and Boehringer Ingelheim also fall into the middle pack, but dedicate smaller shares of their overall pipelines to diseases and countries in scope.

Both Boehringer Ingelheim and Takeda are developing strategies for ensuring that new products for non-communicable diseases (NCDs) reach patients in

relevant countries. As these strategies are not in place yet, most of their NCD products were excluded from this analysis. However, by implementing these strategies, these two companies have substantial potential to improve access to medicine for patients suffering from non-communicable diseases in developing countries.

Laggards have small relevant pipelines

Roche, Pfizer and Eli Lilly lag in product development. Relatively small proportions of their pipelines target diseases relevant to the Index. In addition, Eli Lilly provided little evidence of how its R&D efforts for non-communicable diseases target patients in relevant countries.

Astellas performs poorly in product development due to low disclosure across all parameters. However, it does demonstrate expertise in developing adapted formulations. For example, it is engaged in a partnership to develop a paediatric formulation for treating schistosomiasis.

BEST PRACTICE

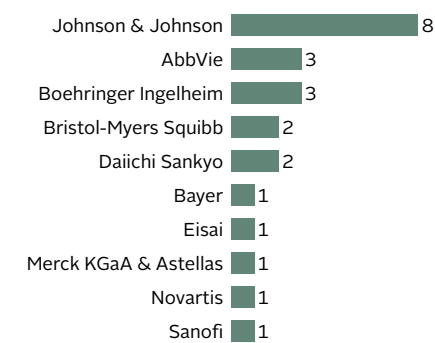
Johnson & Johnson – Adapting products for paediatric use

Children of different ages and weights need differing dosages and dosing forms (such as oral liquids) to ensure effectiveness, safety and compliance.

Of all companies, Johnson & Johnson is adapting the most products for paediatric use that target diseases within the scope of the Index. The company has a strong commitment to targeting unmet needs, and has separate departments for promoting the wellbeing of children: its Janssen Child Health Innovation Leadership Department (CHILD), Paediatric Center of Excellence, Paediatric Advisory Committee and members of its Compound Development teams. Johnson & Johnson is investing approximately USD70 million in 2014 to support paediatric projects of these groups.

- Johnson & Johnson has paediatric medicines in development for HIV/AIDS, hepatitis, diabetes, multi-drug-resistant tuberculosis and soil-transmitted helminthiasis.
- Since 2012, it has gained regulatory approval for a lower-dose tablet for HIV/AIDS and for treating adolescents with schizophrenia with one of its medicines.
- It worked in partnership to ensure its childhood vaccine for meningitis, pertussis, tetanus and hepatitis B could be used in a new delivery device developed for use in remote, resource-limited areas. The device is designed so that community health workers can use it with reduced risk of errors.

Figure 31
Johnson & Johnson is developing most medicines for children



11 companies are developing 23 medicines suited for children.

Partnerships & IP sharing

More R&D partnerships and increased transparency

Why this matters

In recent years, new partnership models have emerged that draw on both public and private funds to pool the risks of R&D and share the benefits.³ As these partnerships become more common, the impact of their terms and conditions on access will also grow.

The Index looks at how and when companies engage in collaborations,

and at whether the underlying R&D contracts include pro-access provisions including, for example, setting price caps, ensuring supply, waiving patent rights, royalty-free provisions and non-exclusive field or territory rights. The Index also examines whether companies have policies for systematically including access provisions while engaging in IP negotiations.

To accelerate early drug-discovery and development, companies can also share their compound libraries and expertise with external parties. The Index looks at whether companies engage in such IP-sharing partnerships to target diseases covered by the Index.

How the companies perform

Companies are developing almost 40% of their relevant products in partnerships. Most of these partnerships target communicable diseases (57%), followed by neglected tropical diseases (21%) and non-communicable diseases (19%). The six conditions targeted most frequently by partnerships are malaria, lower respiratory diseases, HIV/AIDS, hepatitis, diabetes and tuberculosis.

Six companies drive increase in partnerships

Compared with 2012, there has been a 35% increase in the number of relevant product development partnerships and IP-sharing partnerships.^d Nine companies now engage in more partnerships than they did in 2012, with six companies accounting for the majority of the increase; AstraZeneca, Daiichi Sankyo, Eisai, GSK, Novartis and Takeda each engaged in at least five more R&D partnerships than they did in 2012. Conversely, nine companies decreased their number of partnerships since 2012. This may be due to the failure of the compounds in question, to strategic choices or as a result of divestments.

Since the last Index, companies have dramatically increased their transparency around partnerships and their terms and conditions. Overall, more

than half of companies report basing one or more of their product development partnerships on access provisions. Together, these partnerships account for approximately 39% of all product development partnerships. Of the total pipeline, they account for 16% of all product development.

Public partners correlate with access provisions

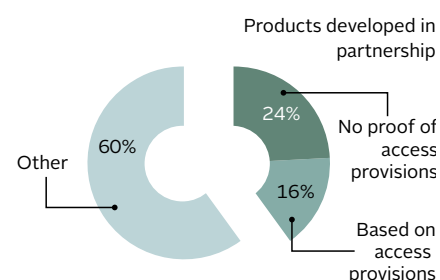
For most communicable diseases, most notably for HIV/AIDS and lower respiratory infections, companies engage with a mix of publicly and privately funded organisations. A large proportion of the private partnerships targeting communicable diseases focus on developing vaccines. Companies provide little evidence that these partnerships are based on access provisions.

For malaria and tuberculosis, companies typically engage in partnerships with publicly funded organisations based on access-oriented terms and conditions. This is also the case for the majority of partnerships targeting neglected tropical diseases.

For non-communicable diseases, there are far fewer partnerships, and companies engage exclusively with other privately funded organisations. Overall,

Figure 32

Almost half of all product development is collaborative



this type of partnership accounts for just 19% of all partnerships. This is to be expected as products for non-communicable diseases typically have functioning markets, at least in developed countries and in the private sector in low income and middle income countries. While this reduces the need for risk sharing via product-development partnerships, it does not guarantee that the resulting products are accessible to the poor or the public sector, especially considering they are often needed for chronic treatment.

Without the involvement of publicly funded organisations, companies are reluctant to disclose the underlying terms and conditions of their

partnership agreements, making it difficult to gauge the accessibility of resulting products.

Leaders agree to access provisions

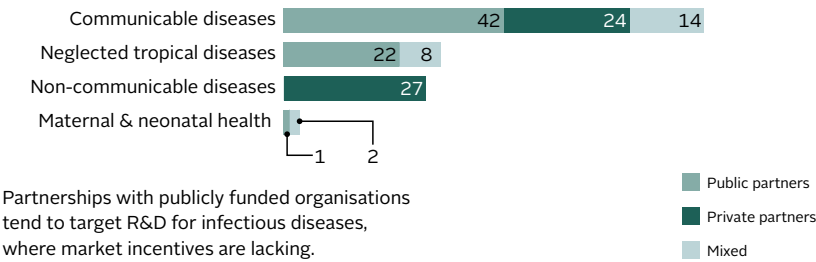
The best-performing companies in partnerships and IP sharing are relatively transparent about the underlying terms and conditions of their R&D contracts, revealing that they regularly include access provisions. As a result, a large proportion of their partnerships are based on access-oriented terms. The leaders also engage in numerous IP-sharing collaborations.

There are eight companies in the leading group: GSK, AbbVie, Merck KGaA, Takeda, Johnson & Johnson, AstraZeneca, Eisai and Novartis.

GSK and AbbVie both develop large proportions of their pipelines in collaboration, and are highly transparent about the related terms and conditions. GSK is the only company that systematically includes access provisions in research contracts for a range of diseases and countries: all projects at its Tres Cantos laboratory in Spain that target tuberculosis, malaria and neglected tropical diseases follow WIPO Re:Search principles, with royalty-free provisions applicable in Least Developed Countries and in materials-transfer or IP-transfer agreements. Other companies can improve by moving in this direction. No company provided evidence of systematically incorporating access provisions into its contracts for a wider range of diseases and countries.

When it comes to IP sharing, AstraZeneca and GSK provided many more external researchers with access to their compound libraries than other companies did. In January 2013, GSK placed information about potential hits for tuberculosis in the public domain, triggering many new research projects. During the period of analysis, AstraZeneca engaged in the most IP sharing through WIPO Re:Search.^a It withdrew from the initiative in September 2013. Seven other companies in the Index

Figure 33
Public partners correlate with greater transparency and access provisions



participate in WIPO Re:Search: GSK, Pfizer, Novartis, Sanofi, Merck & Co., Eisai and Merck KGaA.

Takeda performs well in partnerships and IP sharing, earning it a high position in R&D. It collaborates with three product development partners (PDPs): the TB Alliance, Medicines for Malaria Venture and the Drugs for Neglected Diseases initiative. Takeda is among the leaders when it comes to disclosing the terms and conditions of its partnerships: Takeda joins AbbVie, Merck KGaA, Johnson & Johnson and AstraZeneca in publicly disclosing the access provisions that underpin a number of its partnerships.

Johnson & Johnson, Eisai and Novartis also develop large shares of their overall pipelines in partnership, engage in IP-sharing partnerships and provide evidence of incorporating access provisions in the terms and conditions. However, they did not provide evidence of systematically applying access-oriented principles.

Lack of consistency from middle group
Merck & Co., Gilead, Pfizer, Daiichi Sankyo, Sanofi and Bayer all either develop large proportions of their pipelines in partnership, incorporate access provisions in R&D contracts or engage in numerous IP-sharing partnerships. None of these six companies do all three consistently, and compared with the leaders, they disclose less information about the terms and conditions of their R&D contracts. Merck & Co. stands out for implementing a policy

of systematically incorporating access provisions for certain diseases in its IP agreements.

Roche, Bristol-Myers Squibb and Astellas do not commit to incorporating access provisions, and perform less well in other areas. Roche does not engage in IP sharing, and Bristol-Myers Squibb and Astellas develop just small shares of their pipelines in partnership. Eli Lilly and Bristol-Myers Squibb disclosed only that they engage in public, access-oriented partnerships, instead of disclosing details of the terms and conditions.

The laggards: limited partnerships and less transparency

The laggards are Boehringer Ingelheim and Novo Nordisk. Boehringer Ingelheim reports that it engages in relevant partnerships, but does not disclose the terms and conditions. It also has a comparatively large focus on non-communicable diseases, where there are fewer partnerships and less transparency in general. Its potential lies in ensuring that its in-house developments will be accessible in relevant countries. Novo Nordisk did not report engaging in any relevant R&D partnerships. Drawing on its knowledge and expertise in diabetes care, the company could increase its impact on access to medicine by collaborating with partners with a greater understanding of local needs. Both companies have the potential to increase access to medicine by opening their compound libraries to external researchers looking for potential new hits.

BEST PRACTICE

GSK – Partnering for neonatal health

Drug development targeting neonatal health is scarce.

GSK is one of two companies that is developing medicines for neonatal health conditions. It is the only company working in partnership to develop products expressly meant for use in resource-limited settings. Via its partnership with Save the Children, GSK is developing products for neonatal health conditions with an initial focus on neonatal sepsis. Such partnerships enable companies to match their product development expertise with a partner's knowledge of issues on the ground.



^ Via its PRIME project, GSK trains frontline healthworkers and refurbishes health centers such as this one in DRC.

FOCUS AREA 3

Clinical trial conduct & data transparency

Core group demonstrates higher ethical standards**Why this matters**

Companies are responsible for ensuring that their clinical trials are conducted ethically and to high quality standards. When assessing companies in this regard, the Index refers to two documents: the ICH-GCP standards⁴ for good clinical practice and the Declaration of Helsinki⁵.

ICH-GCP was designed with quality assurance in mind and includes basic ethical considerations, such as procedures for gaining informed consent. The Declaration of Helsinki was designed to ensure patient safety and ethical trial conduct. Compared with ICH-GCP, it includes a larger proportion of ethical considerations.

The Index views compliance with ICH-GCP as a baseline for quality assurance, looking for codes of conduct and procedures for monitoring and auditing that align with its require-

ments. As the Declaration of Helsinki is the gold standard for trial ethics, the Index also assesses whether companies comply with that standard.

Increasingly, clinical trials are conducted in multiple locations and regions,⁶ in countries where regulatory oversight may be weak, and by third-party clinical research organisations (CROs). Working with CROs can save companies time and money while bringing local capacity and expertise on board. However, adding layers of management to global clinical trials raises questions about safety and quality.⁷ Oversight is crucial to prevent misconduct.⁸ The Index reports on companies' processes for monitoring compliance and for taking disciplinary action when necessary, for both in-house and outsourced trials. It also examines companies' criteria for selecting CROs.

By sharing patient-level clinical data, companies can improve the efficiency of product development, improve its cost-effectiveness, enable comparative analyses, and help prevent studies from being repeated unnecessarily.⁹ The industry, patients, healthcare professionals, regulators and the scientific community can all benefit if such data-sharing systems are implemented responsibly, with respect for commercial interests and data confidentiality.

In this regard, the Index looks at whether companies share patient-level data with external partners. The Index also measures companies against the standards set out in the WHO's 2005 Technical Consultation on Clinical Trial Registration Standards.¹⁰ These standards ensure clinical trials are registered centrally, and that their results are published, regardless of outcome.

How the companies perform

Companies generally set high standards for clinical trial conduct and transparency. All comply with ICH-GCP, with seven moving towards compliance with the Declaration of Helsinki (Novartis, Merck & Co., GSK, Merck KGaA, Eli Lilly, Novo Nordisk and Pfizer). These companies all incorporate two or more aspects of the Declaration of Helsinki into their codes of conduct: including ensuring participants can access experimental drugs post-trial; ethical study design considerations; and compensation in case of injury or other serious adverse events. For comparison, in 2012, multiple companies committed to adhering to ethical aspects of the Declaration of Helsinki, generally without providing specifics.

Novartis, Merck & Co., GSK, Boehringer Ingelheim and Johnson & Johnson are the only five companies that provided evidence of having systems in place for sharing patient-level data with external parties and they comply with the WHO standards for trial registration and publication of results. The Index views such developments as an important step forward for healthcare worldwide.

Signs of movement toward higher ethical standard

A few companies approach the higher standards looked for by the Index for enforcing ethical conduct in clinical trials. They provide evidence of extensive auditing and monitoring procedures for ensuring ICH-GCP compliance for both in-house and outsourced trials. They also have disciplinary procedures in place. However, they generally could not demonstrate how they implemented the ethical considerations they draw from the Declaration of Helsinki.

In 2012, the Index reported that only four companies provided evidence of having procedures for enforcing compliance in outsourced trials. In 2014, ten companies provided strong evidence of enforcing compliance through monitoring and auditing procedures for their trials, regardless of

whether they are conducted in-house or outsourced.

When trial results show that a new product is beneficial and safe, companies can provide trial participants with continued access to the study drug until the new product is on the market. While ten companies demonstrate commitment to such post-trial access, none provided evidence of having provided it in practice. The Index encourages companies to improve transparency concerning the implementation of their policies on post-trial access.

The leaders: systems for ensuring high standards of ethical conduct

Novartis, Merck & Co. and GSK lead in this area, with high transparency about their clinical trials and strong clinical trial codes of conduct.

Their clinical trial codes of conduct go beyond compliance with ICH-GCP to incorporate elements of the Declaration of Helsinki. Furthermore, when selecting CROs, they apply criteria designed to ensure quality assurance and quality control in line with ICH-GCP. In case of violations, these companies also have procedures in place for taking disciplinary action.

These top three companies comply with the standards set out by the WHO for registering trials and publishing results regardless of the outcome. In addition, they have systems in place for sharing patient-level clinical study reports (CSRs) with trusted third parties, such as academics.

GSK and Merck & Co. also provided detailed auditing and monitoring information that shows how they enforce ICH-GCP standards in their trials, whether in-house or outsourced. In addition, they provided evidence of having procedures for taking disciplinary action, also for both in-house and outsourced trials. Novartis, however, could extend its auditing and monitoring procedures to outsourced

trials, in order to improve its oversight of CROs.

Middle rankers slip on data transparency and the Declaration of Helsinki

There is a large middle group with high ethical standards for clinical trial conduct, yet varying performances across other parameters. Toward the top of this group are Merck KGaA, Eli Lilly, Novo Nordisk and Pfizer, which incorporate principles unique to the Declaration of Helsinki in their clinical trial codes of conduct. However, they have lower standards than the leaders when it comes to clinical trial data transparency. Following them, Boehringer Ingelheim, Johnson & Johnson, AstraZeneca and Takeda have codes of conduct that comply with ICH-GCP, but show levels of data transparency similar to that of the leaders.

Bayer, Roche and AbbVie all comply with ICH-GCP and have lower standards of clinical trial data transparency than leaders. Takeda, Roche and AbbVie could also be more transparent about their monitoring and auditing procedures.

Lower ethical standards at the bottom

Six companies lag when it comes to clinical trial conduct and transparency: Sanofi, Daiichi Sankyo, Bristol-Myers Squibb, Eisai, Astellas and Gilead. They all have codes of conduct that comply with ICH-GCP, but lack the level of clinical trial transparency of the leaders. In addition, they are missing either policies for taking disciplinary action or criteria for selecting CROs. Bristol-Myers Squibb and Sanofi stand out due to their data sharing policy, and Sanofi has robust oversight mechanisms that extend to CROs. Daiichi Sankyo stands out due to its procedures for selecting CROs and disciplinary action. Conversely, Bristol-Myers Squibb and Gilead can improve the transparency of their monitoring and auditing procedures.

OBSERVATION

Limited evidence of breaches

In general, the industry has endorsed and enforced ICH-GCP compliance as the baseline for clinical trial conduct. While this shows commitment, it doesn't give any indication as to how effective companies' policies, practices and codes of conduct actually are. To provide this, the Index checks whether companies are involved in lawsuits relating to clinical trial conduct in relevant countries or have been issued regulatory notices or fines. Investigations and legal processes can be very lengthy, so the Index reports on rulings and lawsuits from the past five years.

In countries in scope, evidence of breaches of codes of conduct and legislation is scarce and often limited to high-profile cases. It should be noted that the breaches captured by the Index are also a function of the nature and rigour of the various regional and national legal and regulatory frameworks. As such, fewer breaches are likely to be identified in countries with weak reporting systems and judicial controls.

Notices, lawsuits and fines

Two companies received notices from authorities that supervise ethical clinical trial conduct: Bayer and Sanofi. Both are related to clinical trials conducted in India.

- **Bayer** received a notice for not paying compensation following the deaths of clinical trial participants. Providing compensation due to serious adverse events is included in the Declaration of Helsinki. Bayer's code of conduct does not show evidence of these ethical aspects.
- **Sanofi** received a notice for not reporting safety issues that arose during an outsourced trial. Sanofi has mechanisms in place for enforcing ICH-GCP compliance for outsourced trials. This breach indicates that there is still room for misconduct even when enforcement mechanisms are in place.

One company was fined for a breach of clinical trial conduct:

- **GSK** was fined by an Argentinian court in January 2012 for not following procedures for securing informed consent in a trial of a pneumonia vaccine. The trial involved approximately 15,000 babies from

poor rural areas in Latin America, including Colombia. The fine totalled USD220,000.

Three companies are involved in ongoing lawsuits:

- **Merck & Co.** and **GSK** are both involved in a lawsuit in India concerning studies of their cervical cancer vaccines and whether the studies included adequate safety and efficacy tests. The studies were undertaken by PATH, an NGO.
- **Pfizer** is the subject of a lawsuit related to the Trovan trial that took place in Nigeria in 1996 under Wyeth (later acquired by Pfizer). The latest in a series of lawsuits relating to this trial, it relates to alleged breaches of the original settlement.

In addition to these notices, rulings and fines, there are numerous allegations of violations of ethical clinical trial standards, mainly involving study designs that include unethical placebo groups. However, for these cases, no lawsuits have been brought to date.

BEST PRACTICE

Eli Lilly and Daiichi Sankyo – Actively ensuring ethical study designs

The lack of strong regulatory oversight in countries such as India and South Africa creates an environment in which companies and CROs are left to decide whether clinical trial conduct is ethical or not. A few companies are attempting to standardise highly ethical clinical trial conduct by incorporating aspects of the Declaration of Helsinki into their codes of conduct.

Eli Lilly and Daiichi Sankyo go furthest, having put mechanisms in place to ensure study protocols and designs are ethical. Both have established committees for reviewing study protocols for both in-house and outsourced trials. For example, Eli Lilly's review committees routinely ensure that study protocols adhere to its standards and offer an internal consultation service on ethical study design and conduct.

INNOVATION

Innovation characterised by collaboration and partnership

With its high costs and high risks, the traditional R&D model has historically discouraged companies from developing products specifically for diseases that disproportionately affect patients living in developing countries. To fill the resulting gaps, new models for research and development are needed – models that facilitate R&D

efforts for high-burden diseases in these countries while rebalancing the risks and incentives for the companies that adopt them. The Index captures any innovative, sustainable or open business model that addresses current gaps in product development for diseases within the scope of the Index. To qualify for this analysis, these

models must explicitly target the needs of patients living in countries relevant to the Index. The Index also looks for innovative practices that improve the transparency of patient-level data or facilitate ethical clinical trial conduct in countries within the geographic scope of the Index.

Innovation in product development

GSK is investing GBP25 million (USD41 million) to establish its second Open Lab, which will focus on non-communicable diseases in Africa. These diseases are increasingly seen as the next potential public health crisis in developing countries. Research will aim to understand African variations in non-communicable disease types as well as specific treatment needs.¹¹

GHIT is a new multi-stakeholder collaboration co-established by four Japanese companies – Eisai, Takeda, Daiichi Sankyo and Astellas – together with the Japanese government and the Bill & Melinda Gates Foundation.¹² GHIT is a new funding mechanism that provides funding to three product development partnerships (PDPs): the TB Alliance, the Medicines for Malaria Venture and the Drugs for Neglected Diseases Initiative. Through GHIT, these four companies have significantly improved their contributions to R&D for neglected tropical diseases.

TransCelerate is a new joint initiative, launched in 2012, for accelerating and simplifying drug development processes through collaboration.¹³ Twelve of the companies measured by the Index are involved in TransCelerate, which could have an impact on access to medicine, for example, by making the processes related to clinical trials more efficient. The companies involved are: AbbVie, Astellas, AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, Eli Lilly, GSK, Johnson & Johnson, Merck KGaA, Pfizer, Roche and Sanofi.

Innovation in clinical trial conduct and transparency

Eli Lilly has developed a Bioethics Framework that proscribes a higher level of ethical clinical trial conduct than all other companies. It is the only company that supports its clinical trial code of conduct with mechanisms for enforcing the ethical behaviour provisions drawn from the Declaration of Helsinki.

Johnson & Johnson has worked with Yale University to establish an independent external review committee of academics to oversee the sharing of its patient-level data.

For numbered references, see the Appendix.

a WIPO Re:Search is a consortium that aims to build partnerships around companies' intellectual property assets and resources to advance the development of products for neglected tropical diseases, malaria and tuberculosis.

b See the Appendix for the full Access to Medicine Index 2014 Disease Scope.

c See chapter D Pricing, Manufacturing & Distribution.

d Comparison based on analysis criteria from 2012.